

Alexandre Autran

Normalien in Mathematics and Machine Learning

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Profile

I am a machine learning and mathematics student at the École Normale Supérieure (ENS) in Paris-Saclay and Rennes. My research interests are centered on probabilistic and geometric methods for machine learning, high-dimensional statistics, density analysis, optimal transport, graph-based learning, and statistical learning theory.

Education

2024–2027 ENS Diploma, Department of Mathematics

Admitted to the ENS through a competitive entrance examination.

2026–2027 Research year in machine learning, École Normale Supérieure Paris-Saclay, Paris, France

Final year of the ENS Diploma (research track at ENS de Rennes).

2025–2027 MSc in Mathematics, École Normale Supérieure de Rennes, France

MSc in Mathematics (M2). Highest honors (17/20), first semester.

MSc in Mathematics (M1). Highest honors (17.6/20). Rank: 5th/60+.

2024–2025 BSc in Mathematics, Computer Science minor, École Normale Supérieure de Rennes, France

High honors (15.4/20).

Publications

2026 **A. Autran**, “Second-Order Asymptotics of Emergent α -Stable Common Noise”, preprint.

2026 **A. Autran**, “Optimization with an Exact Discrete Adjoint for a Diffusive Lotka–Volterra Harvesting Problem”, preprint.

2026 **A. Autran**, D. Munteanu, and J.-L. Autran, “Multi-Dimensional Charge Diffusion–Collection Model in Semiconductor Devices Subjected to Single Ionizing Particles”, *Applied Sciences*, vol. 16, no. 11, Art. no. 5551, 2026. [[Paper](#)]

Research Experience

Aug 2026– Machine Learning Researcher

High-Dimensional Density Analysis Using Local Unimodality.

Supervised by Argyris Kalogeratos (Senior Researcher) at Centre Borelli, ENS Paris-Saclay, France.

May–Aug 2026 MSc Research Internship

Optical Modeling of Thermochromic VO₂-Based Oxide Nanocomposites [[Report](#)] [[Presentation](#)]

Supervised by Prof. Nguyen Ngoc Duy and Amaury Baret (PhD Student) at the University of Liège, Belgium.

Apr–Jun 2026 MSc Research Internship

Numerical Simulation and Optimal Control of Spatial Lotka–Volterra Systems. [[Report](#)]

Supervised by Prof. Martin J. Gander and Bastien Chaudet-Dumas (Postdoctoral Researcher) at the University of Geneva, Switzerland.

Apr–May 2026 Machine Learning Projects

Cardiovascular Disease Prediction Using Statistical Learning and Neural Network Models; Air Pollution Prediction from Environmental Data. [[Cardiovascular report](#)] [[Air-pollution report](#)]

Studies carried out as part of the MSc course *Statistical Learning* (Prof. Myriam Maumy).

Jan 2026 Statistical Reading Group

Supervised Statistical Learning – Prediction by Empirical Risk Minimization. [[Presentation](#)]

Supervised by Prof. Magalie Fromont at ENS de Rennes, France.

Nov–Dec 2025 MSc Research Seminar

Point Process and Stein’s Method. [[Report](#)] [[Presentation](#)]

Supervised by Prof. Jean-Christophe Breton at the University of Rennes, France.

- Oct–Nov 2025 **Algebra Reading Group**
Elimination Theory and Extension. [[Presentation](#)]
Supervised by Jade Nardi (Research Scientist) at *ENS de Rennes*, France.
- Jun–Aug 2025 **BSc Research Internship**
Lattice-Based Post-Quantum Cryptography. [[Report](#)] [[Presentation](#)]
Supervised by Prof. Pierre-Alain Fouque and Aurore Guillevic (Research Scientist) at Inria, France.
- May–Jun 2025 **BSc Research Internship**
Algebraic Counting of the Real Roots of Polynomials. [[Report](#)] [[Presentation](#)]
Supervised by Prof. Marie-Françoise Roy at the University of Rennes, France.
- Feb–Mar 2025 **Directed Research Reading**
Geometry of Numbers. [[Report](#)] [[Presentation](#)]
Supervised by Juliette Bavard (Research Scientist) at ENS de Rennes, France.
- Feb–Mar 2024 **BSc Research Internship**
Topological Study of Matrix Groups. [[Report](#)]
Supervised by Prof. Françoise Dal'Bo and Jérémy Bettinger (PhD Student) at the University of Rennes, France.
- May–Jun 2023 **BSc Research Internship**
Introduction to Hyperbolic Geometry and Fuchsian Groups. [[Report](#)] [[Presentation](#)]
Supervised by Frank Loray (CNRS Research Director) at the University of Rennes, France.

Other Activities

- 2024–2025 **ENS Oral Examinations**
Supervision of 2nd year BSc students in preparation for the oral entrance examinations in mathematics of the Écoles Normales Supérieures: Ulm, Paris-Saclay, Rennes and Lyon. [[Resources](#)]
- 2022–2025 **Teaching in Mathematics**
Tutoring for final-year high school students of the Lycée Chateaubriand in Rennes, and for first and second years BSc students of the University of Rennes. [[Resources](#)]
- 2023–2024 **BSc Student Representative**
Representative on the council of the Mathematics Department of the University of Rennes.
- Before 2022 **High School Competitions**
Concours Général in Mathematics, Physics, Philosophy, and National High School Mathematics Olympiad at Lycée Thiers in Marseille.

Computer Skills

Programming: Python, C/C++, Java, MATLAB, R.
Python / ML: NumPy, pandas, scikit-learn, PyTorch.

Languages

French: native language. **English:** professional working proficiency.

Sports and Interests

Sports: running, cycling, strength training.
Interests: reading and writing, in fantasy and science fiction literature; chess.

Academic Reference

François Bolley [[Web page](#)]
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